

Alexander Frank

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SUMMARY

Software engineer, 5+ years building mission-critical crew scheduling systems at Southwest Airlines that operational teams depend on daily. Full-stack in production across TypeScript, React, Node.js, and Kubernetes; independent work in Rust systems programming and parallel developer tooling.

TECHNICAL SKILLS

Languages: TypeScript, JavaScript, Java, Python, Rust, SQL, Bash

Frontend: React, Next.js, HTML5, CSS3, SASS

Backend & Cloud: Node.js, GraphQL, REST APIs, business rules engines, AWS, Docker, Kubernetes, CI/CD, Linux, Git

AI & Agents: Claude Agent SDK, LLM API integration, multi-agent orchestration

PROFESSIONAL EXPERIENCE

Southwest Airlines

June 2021 – Present

Software Engineer — Crew Scheduling & Recovery

- Build and maintain the mission-critical scheduling applications that 25,000+ crew members (pilots and flight attendants) and hundreds of operational schedulers depend on daily
- Cut a recurring 5-minute manual log-retrieval workflow to under 5 seconds (98% faster) by automating Kubernetes log collection in Bash, shortening incident triage for the whole team
- Enabled other engineers to independently develop and maintain complex scheduling logic and decision rules by leading knowledge-transfer sessions on the team's business rule management system
- Mentor associate-level engineers through onboarding and technical growth; active member of Southwest's College Hire and Intern Program (CHIP)

PROJECTS

Sunrise

TypeScript · React · Node.js · Claude Agent SDK

- Developer tool that turns a markdown plan into parallel execution: one agent per task heading, each running in its own git worktree and branch so concurrent tasks cannot conflict
- Local web cockpit streams live task state and surfaces any task blocked on human input, keeping a parallel run reviewable and interruptible from one screen

Reflect

github.com/alexfrankcodes/reflect

Rust · Tauri · Windows

- Windows tray journaling app built local-first: every entry is a plain text file on the user's own machine, so data stays portable and fully offline with no account or server
- Chose a native Rust core packaged with Tauri over an Electron shell, trading ecosystem breadth for a small binary and low idle memory in an always-resident tray process

EDUCATION

San Jose State University

Bachelor of Science, Software Engineering